EC303 — Signals and System

Semester III • 3L:0T:0P • 3 Credits

Module 4 Exam Notes: Laplace Transform, Z-Transform & LSI System Analysis

CGEC | B.Tech ECE (2023–27) | *Biraj Sarkar*

Contents

1 Evolution of Integral Transforms **2**
- 1.1 Limitations of the Fourier Transform 2
- 1.2 Generalization to the Laplace Transform (Continuous Domain) 2
- 1.3 Generalization to the Z-Transform (Discrete Domain) 2

2 The Laplace Transform **3**
- 2.1 Bilateral and Unilateral Laplace Transforms 3
- 2.2 Eigenfunctions and Eigenfunction Basis 3
- 2.3 Region of Convergence (ROC) for Laplace Transform 4
- 2.4 System Causality and Stability via ROC 4
- 2.5 Poles, Zeros, and Transfer Function $H(s)$ 5
- 2.6 Solution of Differential Equations via Unilateral Laplace Transform 5

3 The Z-Transform for Discrete-Time Signals and Systems **5**
- 3.1 Bilateral and Unilateral Z-Transforms 5
- 3.2 Properties of the Region of Convergence (ROC) in the z-Plane 6
- 3.3 Causality and Stability of DT LSI Systems 6
- 3.4 Common Z-Transform Pairs 7
- 3.5 Z-Domain Analysis of Difference Equations 7

1. Evolution of Integral Transforms

1.1 Limitations of the Fourier Transform

The Continuous-Time Fourier Transform (CTFT) requires Dirichlet conditions to converge, fundamentally demanding that the signal be absolutely integrable:

$$\int_{-\infty}^{\infty} |x(t)|dt < \infty \tag{1}$$

Consequently, many vital engineering signals cannot be transformed directly using standard CTFT without resorting to distributions or impulse trains:

1. Growing exponentials: $x(t) = e^{at}u(t)$ with $a > 0$ (unstable system responses).
2. Signals of infinite energy and non-decaying nature (e.g., polynomial ramps $t^n u(t)$).
3. Systems with non-zero initial conditions (differential equations with initial stored energy).

1.2 Generalization to the Laplace Transform (Continuous Domain)

To ensure convergence for signals that grow or do not decay, we introduce an exponential damping factor $e^{-\sigma t}$ ($\sigma \in \mathbb{R}$) prior to computing the Fourier transform:

$$\mathcal{L}\{x(t)\} = \mathcal{F}\{x(t)e^{-\sigma t}\} = \int_{-\infty}^{\infty} [x(t)e^{-\sigma t}]e^{-j\omega t}dt \tag{2}$$

$$= \int_{-\infty}^{\infty} x(t)e^{-(\sigma+j\omega)t}dt \tag{3}$$

Defining the **complex frequency** variable $s \triangleq \sigma + j\omega \in \mathbb{C}$, we arrive directly at the **Bilateral Laplace Transform**:

$$X(s) = \int_{-\infty}^{\infty} x(t)e^{-st}dt \tag{4}$$

The Fourier transform is simply the Laplace transform evaluated along the imaginary axis ($s = j\omega$, i.e., $\sigma = 0$):

$$X(j\omega) = X(s)\big|_{s=j\omega} \quad \text{(provided the } j\omega\text{-axis lies within the ROC)} \tag{5}$$

1.3 Generalization to the Z-Transform (Discrete Domain)

Similarly, the Discrete-Time Fourier Transform (DTFT) requires absolute summability $\sum |x[n]| < \infty$. By applying a radial damping factor r^{-n} ($r > 0$) to $x[n]$:

$$\mathcal{Z}\{x[n]\} = \text{DTFT}\{x[n]r^{-n}\} = \sum_{n=-\infty}^{\infty} (x[n]r^{-n})e^{-j\Omega n} \tag{6}$$

$$= \sum_{n=-\infty}^{\infty} x[n](re^{j\Omega})^{-n} \tag{7}$$

Defining the complex variable $z \triangleq re^{j\Omega} \in \mathbb{C}$, we obtain the **Bilateral Z-Transform**:

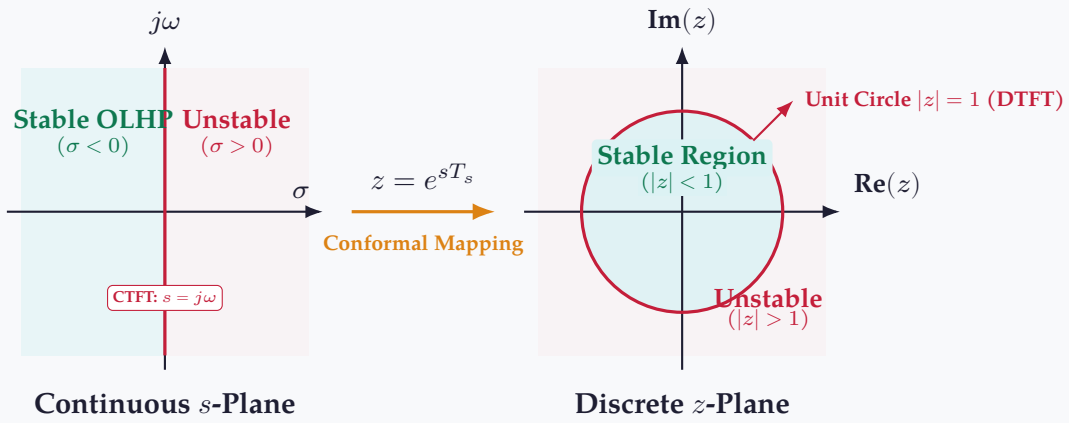
$$X(z) = \sum_{n=-\infty}^{\infty} x[n]z^{-n} \tag{8}$$

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The DTFT is simply the Z-transform evaluated on the **unit circle** ($|z| = r = 1$, i.e., $z = e^{j\Omega}$):

$$X(e^{j\Omega}) = X(z)|_{z=e^{j\Omega}} \quad \text{(provided the unit circle lies within the ROC)} \tag{9}$$

TikZ Schematic: Mapping Between s-Plane and z-Plane ($z = e^{sT_s}$)



2. The Laplace Transform

2.1 Bilateral and Unilateral Laplace Transforms

Definition: Bilateral vs Unilateral Laplace Transform

$$\textbf{Bilateral (Two-Sided):} \quad X(s) = \mathcal{L}\{x(t)\} \triangleq \int_{-\infty}^{\infty} x(t)e^{-st}dt \tag{10}$$

$$\textbf{Unilateral (Single-Sided):} \quad X(s) = \mathcal{L}_u\{x(t)\} \triangleq \int_{0^-}^{\infty} x(t)e^{-st}dt \tag{11}$$

where 0^- includes impulses located at the origin $t = 0$.

Why Unilateral Transform is Crucial for Engineers:

- Physical systems are causal, and experiments start at $t = 0$.
- It naturally incorporates non-zero initial conditions $(y(0^-), y'(0^-))$, enabling algebraic solution of differential equations with initial stored energy.

2.2 Eigenfunctions and Eigenfunction Basis

As established in Module 3, $x(t) = e^{st}$ is an eigenfunction of an LSI system:

$$y(t) = h(t) * e^{st} = \left[\int_{-\infty}^{\infty} h(\tau)e^{-s\tau}d\tau\right] e^{st} = H(s) \cdot e^{st} \tag{12}$$

The transfer function $H(s)$ is the eigenvalue. The continuum of eigenfunctions $\{e^{st}\}_{s\in\text{ROC}}$ forms a complete mathematical basis spanning the signal space of Laplace-transformable signals.

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2.3 Region of Convergence (ROC) for Laplace Transform

Definition: Region of Convergence (ROC)

The **Region of Convergence (ROC)** is the set of complex values of $s = \sigma + j\omega$ in the s-plane for which the Laplace integral converges absolutely:

$$\text{ROC} = \left\{ s \in \mathbb{C} : \int_{-\infty}^{\infty} |x(t)| e^{-\sigma t} dt < \infty \right\} \tag{13}$$

Fundamental Properties of the Laplace ROC:

1. **Strip Geometry:** The ROC consists of vertical strips parallel to the $j\omega$-axis because convergence depends solely on $\sigma = \text{Re}(s)$, not on $\omega = \text{Im}(s)$.
2. **Poles Exclusion:** The ROC contains **no poles** (since $X(s) \to \infty$ at poles).
3. **Finite Duration Signals:** If $x(t)$ is non-zero only for $T_1 \leq t \leq T_2$ and is absolutely integrable, the ROC is the **entire s-plane**.
4. **Right-Sided Signals:** If $x(t)$ is right-sided ($x(t) = 0$ for $t < T_1$), the ROC is a right-half plane:
$$\text{Re}(s) > \sigma_{\text{max}} = \max_k \{\text{Re}(p_k)\} \tag{14}$$
5. **Left-Sided Signals:** If $x(t)$ is left-sided ($x(t) = 0$ for $t > T_2$), the ROC is a left-half plane:
$$\text{Re}(s) < \sigma_{\text{min}} = \min_k \{\text{Re}(p_k)\} \tag{15}$$
6. **Two-Sided Signals:** If $x(t)$ is two-sided, the ROC is an open vertical strip: $\sigma_1 < \text{Re}(s) < \sigma_2$.

2.4 System Causality and Stability via ROC

Causality and BIBO Stability Criteria in the s-Plane

1. **Causality Criterion:** An LTI system is causal if and only if its ROC is a **right-half plane** extending to the right of the rightmost pole:
$$\text{ROC}: \quad \text{Re}(s) > \max_k \{\text{Re}(p_k)\} \tag{16}$$
2. **BIBO Stability Criterion:** An LTI system is BIBO stable if and only if the **ROC includes the imaginary axis** ($j\omega$-axis, $\text{Re}(s) = 0$).
3. **Causal + Stable Systems:** For an LTI system to be simultaneously causal and stable, **all poles must lie strictly in the Open Left-Half of the s-plane (OLHP)**:
$$\text{Re}(p_k) < 0 \quad \text{for all poles } p_k \tag{17}$$

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2.5 Poles, Zeros, and Transfer Function $H(s)$

The transfer function of a rational continuous-time LSI system is:

$$H(s) = \frac{Y(s)}{X(s)} = \frac{b_M s^M + b_{M-1}s^{M-1} + \cdots + b_0}{a_N s^N + a_{N-1}s^{N-1} + \cdots + a_0} = K\frac{\prod_{i=1}^{M}(s - z_i)}{\prod_{k=1}^{N}(s - p_k)} \tag{18}$$

where:

- z_i are the **Zeros** (frequencies where $H(s) = 0$).
- p_k are the **Poles** (frequencies where $H(s) \to \infty$).
- **Frequency Response Evaluation from Pole-Zero Vectors:**

$$|H(j\omega)| = |K|\frac{\prod_{i=1}^{M}|j\omega - z_i|}{\prod_{k=1}^{N}|j\omega - p_k|}, \qquad \angle H(j\omega) = \sum_{i=1}^{M}\angle(j\omega - z_i) - \sum_{k=1}^{N}\angle(j\omega - p_k) \tag{19}$$

2.6 Solution of Differential Equations via Unilateral Laplace Transform

The unilateral Laplace transform differentiation theorems state:

$$\mathcal{L}_u\left\{\frac{dy(t)}{dt}\right\} = sY(s) - y(0^-) \tag{20}$$

$$\mathcal{L}_u\left\{\frac{d^2y(t)}{dt^2}\right\} = s^2Y(s) - sy(0^-) - y'(0^-) \tag{21}$$

Applying this to an N-th order LCCDE yields:

$$A(s)Y(s) - I(s) = B(s)X(s) \implies Y(s) = \underbrace{\frac{I(s)}{A(s)}}_{Y_{zi}(s)\text{ (ZIR)}} + \underbrace{\frac{B(s)}{A(s)}X(s)}_{Y_{zs}(s)\text{ (ZSR)}} \tag{22}$$

where $I(s)$ contains all initial condition terms, and $H(s) = B(s)/A(s)$. Taking the inverse Laplace transform via Partial Fraction Expansion (PFE) yields the complete time-domain response $y(t) = y_{zi}(t) + y_{zs}(t)$.

3. The Z-Transform for Discrete-Time Signals and Systems

3.1 Bilateral and Unilateral Z-Transforms

Definition: Bilateral vs Unilateral Z-Transform

$$\textbf{Bilateral (Two-Sided):} \quad X(z) = \mathcal{Z}\{x[n]\} \triangleq \sum_{n=-\infty}^{\infty} x[n]z^{-n} \tag{23}$$

$$\textbf{Unilateral (Single-Sided):} \quad X(z) = \mathcal{Z}_u\{x[n]\} \triangleq \sum_{n=0}^{\infty} x[n]z^{-n} \tag{24}$$

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Eigenfunctions in Discrete-Time: The geometric sequence $x[n] = z^n$ ($z \in \mathbb{C}$) is the eigenfunction of DT LSI systems:

$$y[n] = h[n] * z^n = \sum_{k=-\infty}^{\infty} h[k] z^{n-k} = z^n \left[\sum_{k=-\infty}^{\infty} h[k] z^{-k} \right] = H(z) \cdot z^n \tag{25}$$

The eigenvalue is the discrete system transfer function $H(z)$.

3.2 Properties of the Region of Convergence (ROC) in the z-Plane

Definition: ROC in the z-Plane

The ROC is the set of points in the complex z-plane for which the Laurent series converges absolutely:

$$\text{ROC} = \left\{ z \in \mathbb{C} : \sum_{n=-\infty}^{\infty} |x[n]| \cdot |z|^{-n} < \infty \right\} \tag{26}$$

Core Geometric Properties:

1. **Annular Rings:** The ROC is an open circular ring centered at the origin: $r_1 < |z| < r_2$.
2. **No Poles in ROC:** The ROC cannot contain any poles.
3. **Finite Length Sequences:** The ROC is the entire z-plane except possibly $z = 0$ (if $x[n]$ has non-zero values for $n > 0$) and/or $z = \infty$ (if $x[n]$ has non-zero values for $n < 0$).
4. **Right-Sided Sequences:** If $x[n]$ is right-sided ($x[n] = 0$ for $n < N_1$), the ROC is the exterior of a circle:
$$|z| > r_{\text{max}} = \max_k |p_k| \tag{27}$$
5. **Left-Sided Sequences:** If $x[n]$ is left-sided ($x[n] = 0$ for $n > N_2$), the ROC is the interior of a circle:
$$|z| < r_{\text{min}} = \min_k |p_k| \tag{28}$$
6. **Two-Sided Sequences:** The ROC is an open annular ring between two concentric circles: $r_1 < |z| < r_2$.

3.3 Causality and Stability of DT LSI Systems

Causality and BIBO Stability Criteria in the z-Plane

1. **Causality Criterion:** A discrete-time LTI system is causal if and only if its ROC is the **exterior of a circle** outside the outermost pole, including $z = \infty$:
$$\text{ROC}: \quad |z| > \max_k |p_k| \tag{29}$$
2. **BIBO Stability Criterion:** A discrete-time LTI system is BIBO stable if and only if the **ROC includes the unit circle** ($|z| = 1$).
3. **Causal + Stable Systems:** For a DT system to be simultaneously causal and stable, **all poles must lie strictly inside the unit circle**:
$$|p_k| < 1 \quad \text{for all poles } p_k \tag{30}$$

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3.4 Common Z-Transform Pairs

Sequence $x[n]$	**Z-Transform** $X(z)$	**ROC**
$\delta[n]$	1	Entire z-plane
$u[n]$	$\frac{1}{1-z^{-1}} = \frac{z}{z-1}$	$\lvert z\rvert > 1$
$-u[-n-1]$	$\frac{1}{1-z^{-1}} = \frac{z}{z-1}$	$\lvert z\rvert < 1$
$a^n u[n]$	$\frac{1}{1-az^{-1}} = \frac{z}{z-a}$	$\lvert z\rvert > \lvert a\rvert$
$-a^n u[-n-1]$	$\frac{1}{1-az^{-1}} = \frac{z}{z-a}$	$\lvert z\rvert < \lvert a\rvert$
$na^n u[n]$	$\frac{az^{-1}}{(1-az^{-1})^2} = \frac{az}{(z-a)^2}$	$\lvert z\rvert > \lvert a\rvert$
$\cos(\Omega_0 n)u[n]$	$\frac{1-\cos(\Omega_0)z^{-1}}{1-2\cos(\Omega_0)z^{-1}+z^{-2}}$	$\lvert z\rvert > 1$

3.5 Z-Domain Analysis of Difference Equations

The unilateral Z-transform time-delay properties incorporate initial conditions:

$$\mathcal{Z}_u\{y[n-1]\} = z^{-1}Y(z) + y[-1] \tag{31}$$
$$\mathcal{Z}_u\{y[n-2]\} = z^{-2}Y(z) + z^{-1}y[-1] + y[-2] \tag{32}$$

Applying these to an N-th order difference equation $\sum_{k=0}^{N} a_k y[n-k] = \sum_{m=0}^{M} b_m x[n-m]$ transforms difference equations into algebraic equations in z, decomposing into Zero-Input Response (ZIR) and Zero-State Response (ZSR).

Worked Example: Complete Response via Unilateral Z-Transform

Solve the discrete difference equation $y[n] - 0.5y[n-1] = x[n]$ for $n \geq 0$ with initial condition $y[-1] = 2$ and unit step excitation $x[n] = u[n]$:

Step 1: Apply Unilateral Z-Transform: $Y(z) - 0.5[z^{-1}Y(z) + y[-1]] = X(z) = \frac{1}{1-z^{-1}}$.

Step 2: Substitute $y[-1] = 2$ and Decompose Responses:

$$Y(z)(1-0.5z^{-1}) - 1 = \frac{1}{1-z^{-1}} \implies Y(z) = \underbrace{\frac{1}{1-0.5z^{-1}}}_{Y_{zi}(z)\text{ (ZIR)}} + \underbrace{\frac{1}{(1-0.5z^{-1})(1-z^{-1})}}_{Y_{zs}(z)\text{ (ZSR)}}$$

Step 3: Partial Fraction Expansion of Zero-State Component:

$$Y_{zs}(z) = \frac{A}{1-z^{-1}} + \frac{B}{1-0.5z^{-1}} \implies A = 2, \quad B = -1$$

Taking the inverse Z-transform yields:

- **Zero-Input Response (ZIR):** $y_{zi}[n] = \mathcal{Z}^{-1}\left\{\frac{1}{1-0.5z^{-1}}\right\} = (0.5)^n u[n]$.
- **Zero-State Response (ZSR):** $y_{zs}[n] = [2 - (0.5)^n]\, u[n]$.

Step 4: Total Complete System Response:

$$y[n] = y_{zi}[n] + y_{zs}[n] = (0.5)^n u[n] + [2 - (0.5)^n]\, u[n] = \mathbf{2u[n]} \quad (n \geq 0)$$

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Quick Revision — Module 4

Laplace Transform, Z-Transform & LSI System Analysis | EC303

Key Formulas at a Glance

Bilateral Laplace Transform	$X(s) = \int_{-\infty}^{\infty} x(t)e^{-st}dt, \quad s = \sigma + j\omega$
Unilateral Laplace Transform	$X(s) = \int_{0^-}^{\infty} x(t)e^{-st}dt$
Laplace Derivative Property	$\mathcal{L}_u\{\dot{y}\} = sY(s) - y(0^-), \quad \mathcal{L}_u\{\ddot{y}\} = s^2Y - sy(0^-) - y'(0^-)$
Continuous Causal+Stable Criterion	All poles in Open Left-Half Plane: $\text{Re}(p_k) < 0$
Continuous Transfer Function	$H(s) = \mathcal{L}\{h(t)\} = Y(s)/X(s)$
Bilateral Z-Transform	$X(z) = \sum_{n=-\infty}^{\infty} x[n]z^{-n}, \quad z = re^{j\Omega}$
Unilateral Z-Transform	$X(z) = \sum_{n=0}^{\infty} x[n]z^{-n}$
Z-Transform Delay Property	$\mathcal{Z}_u\{y[n-1]\} = z^{-1}Y(z) + y[-1]$
Discrete Causal+Stable Criterion	All poles inside Unit Circle: $\lvert p_k\rvert < 1$
DTFT $\leftrightarrow$ Z-Transform	$X(e^{j\Omega}) = X(z)\big\rvert_{z=e^{j\Omega}}$ (if unit circle in ROC)

One-Line Definitions

- **Laplace Transform:** A generalized Fourier transform with real damping factor $e^{-\sigma t}$ extending convergence to unstable signals.
- **Z-Transform:** A generalized discrete-time Fourier transform with radial damping factor r^{-n} extending convergence.
- **Region of Convergence (ROC):** The domain in the complex plane (s or z) where the transform integral/series converges absolutely.
- **Poles of a System:** Complex frequencies at which the transfer function denominator vanishes, causing infinite system gain.
- **Zeros of a System:** Complex frequencies at which the transfer function numerator vanishes, completely blocking transmission.
- **Transfer Function $H(s)/H(z)$:** The ratio of the transform of the output to the transform of the input under zero initial conditions.
- **Open Left-Half Plane (OLHP):** The region $\text{Re}(s) < 0$ where all continuous-time system poles must lie for BIBO stability.
- **Unit Circle ($|z| = 1$):** The contour in the z-plane corresponding to the DTFT and the boundary of discrete-time BIBO stability.

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- **Unilateral Transform:** A transform defined strictly over non-negative time, tailored for causal systems and initial value problems.

Frequently Asked / Viva Questions

Q1. Why can the Fourier transform not handle the unit step function $u(t)$ directly without impulses?

Because $u(t)$ is not absolutely integrable ($\int_0^\infty 1dt = \infty$). The convergence integral diverges on the imaginary axis. The Laplace transform introduces the factor $e^{-\sigma t}$ ($\sigma > 0$), forcing the integral to converge to $1/s$ in the ROC $\text{Re}(s) > 0$.

Q2. Can two distinct time-domain signals have identical Laplace or Z-transform expressions?

Yes. For example, $x_1(t) = e^{-at}u(t)$ and $x_2(t) = -e^{-at}u(-t)$ both have the algebraic expression $1/(s+a)$. They are distinguished solely by their ROCs: $\text{Re}(s) > -a$ for $x_1(t)$ and $\text{Re}(s) < -a$ for $x_2(t)$. A transform is mathematically incomplete without specifying its ROC.

Q3. State the condition on the ROC for an LTI system to be both causal and BIBO stable in continuous time.

The ROC must be a right-half plane $\text{Re}(s) > \sigma_{\text{max}}$ (for causality), and it must contain the imaginary axis $\text{Re}(s) = 0$ (for stability). This implies that $\sigma_{\text{max}} < 0$, meaning all poles must lie strictly in the Open Left-Half of the s-plane.

Q4. What is the geometric mapping relationship between the s-plane and the z-plane?

The mapping is defined by $z = e^{sT_s}$. The imaginary axis $\text{Re}(s) = 0$ maps to the unit circle $|z| = 1$. The stable Open Left-Half Plane $\text{Re}(s) < 0$ maps to the interior of the unit circle $|z| < 1$. The unstable Right-Half Plane $\text{Re}(s) > 0$ maps to the exterior $|z| > 1$.

Q5. What is the condition on poles for a causal discrete-time LTI system to be BIBO stable?

All poles must lie strictly inside the unit circle of the z-plane ($|p_k| < 1$ for all k).

Q6. Why does the unilateral Laplace transform include initial conditions while the bilateral transform does not?

Because the integration begins at $t = 0^-$, the boundary values $y(0^-), y'(0^-)$ naturally emerge via integration by parts when evaluating derivatives, capturing internal stored energy.

Q7. What is the physical interpretation of system zeros?

Zeros are frequencies where the system transmission is zero. If an input phasor has a frequency matching a zero ($s = z_i$), the steady-state output is completely extinguished.

Q8. Why can a finite-duration signal have an ROC that is the entire complex plane?

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Frequently Asked / Viva Questions (contd.)

For a signal of finite duration $[T_1, T_2]$, the integration/summation contains only finite limits. Provided the signal does not contain singularities of infinite area, the integral of $|x(t)|e^{-\sigma t}$ over a compact interval is always finite for any real σ.

Q9. How do you determine the steady-state frequency response $H(j\omega)$ geometrically from the pole-zero plot?

At any point $j\omega$ on the imaginary axis, draw vectors from all zeros and poles to $j\omega$. The magnitude $|H(j\omega)|$ is the product of lengths of zero-vectors divided by the product of lengths of pole-vectors (scaled by gain K). The phase is the sum of angles of zero-vectors minus the sum of angles of pole-vectors.

Q10. What is the Initial Value Theorem in the Laplace domain?

If $x(t)$ and its derivative are Laplace transformable and $x(t) = 0$ for $t < 0$, then $\lim_{t\to 0^+} x(t) = \lim_{s\to\infty} sX(s)$ (provided the limit exists).

Summary Matrix: Comprehensive Comparison of Transforms

Transform	Kernel / Variable	Domain & ROC	Stability Criterion
CTFT	$e^{-j\omega t}$, $\omega \in \mathbb{R}$	**Imaginary axis**	Requires $\int \|x(t)\|dt < \infty$.
Laplace (s)	e^{-st}, $s = \sigma + j\omega$	**Vertical strips in s-plane**	ROC includes $j\omega$-axis (poles in OLHP).
DTFT	$e^{-j\Omega n}$, $\Omega \in [-\pi, \pi]$	**Unit circle ($\|z\| = 1$)**	Requires $\sum \|x[n]\| < \infty$.
Z-Transform (z)	z^{-n}, $z = re^{j\Omega}$	**Annular rings in z-plane**	ROC includes unit circle (poles inside $\|z\| < 1$).

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